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EXAMPLE 4

Geometric Mean Return

1. An analyst observes the following annual rates of return for a hedge fund, which are presented in Exhibit 4.

Exhibit 4: Hedge Fund Performance, 20X8–20X0

Year	Return (%)
20X8	22
20X9	–25
20X0	11

The fund's geometric mean return over the three-year period is *closest* to:

A. 0.52 percent.

B. 1.02 percent.

C. 2.67 percent.

Solution:

A is correct. Applying Equation 4, the fund's geometric mean return over the three-year period is 0.52 percent, calculated as follows:

$$\overline{R}_G = [(1 + 0.22)(1 - 0.25)(1 + 0.11)]^{(1/3)} - 1 = 1.0157^{(1/3)} - 1 = 0.0052$$
$$= 0.52\%.$$

EXAMPLE 5

Geometric and Arithmetic Mean Returns

1. Consider the annual return data for the group of countries in Exhibit 5.

Exhibit 5: Annual Returns for Years 1 to 3 for Selected Countries' Stock Indexes

	52-Week Return (%)			Average 3-Year Return	
Index	Year 1	Year 2	Year 3	Arithmetic	Geometric
Country A	–15.6	–5.4	6.1	–4.97	–5.38
Country B	7.8	6.3	–1.5	4.20	4.12
Country C	5.3	1.2	3.5	3.33	3.32
Country D	–2.4	–3.1	6.2	0.23	0.15
Country E	–4.0	–3.0	3.0	–1.33	–1.38
Country F	5.4	5.2	–1.0	3.20	3.16
Country G	12.7	6.7	–1.2	6.07	5.91
Country H	3.5	4.3	3.4	3.73	3.73
Country I	6.2	7.8	3.2	5.73	5.72